OPENSEARCH IN AWS

A TRADE STUDY - FEBRUARY 2026

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Objectives

1. Compare and contrast **Amazon OpenSearch Service** (AOSS) vs **AWS Managed OpenSearch**
   - Self-managed OpenSearch is beyond scope of consideration due to complexity and administrative overhead - solution offers the highest degree of control but at a massive cost of effort.
2. Characterize the PDS OpenSearch Use-Case
   - Storage
     - Baseline Amount
     - Ingest Trend(s)
   - Search/queries
     - Workload Behavior(s)
     - Activity Pattern(s)
   - Depict Current Deployment
3. Apply #1 to suggest the optimal solution for #2.

AOSS

- OCU - OpenSearch Compute Unit
  - 1 vCPU, 8GB RAM, 120 GB storage
- "Serverless" (is it really though?)
  - automagically scales (up and down) # of OCUs based on:
    - index: ingest activity
    - search: volume and query activity
    - redundancy/HA : 2x index OCUs
- Collections
  - Set of indexes
  - Types: search, vector, time series
- Capacity limits: # of OCUs
  - maximum: point at which excessive requests are rejected

AOSS

Costs:

- Compute: $0.24 / OCU-hour
  - No notion of reserved OCUs
- Storage: $0.024 / GB-month
  - Search collections: exclusively Hot storage

AOSS

Pros:

- Minimal manual management req'd
  - Capacity limits
- Scaling, sharding, etc all performed by the service

Cons:

- Mandatory minimums
  - Storage : 2 OCUs / 120GB storage
    - Further impacted by shards, # of indexes

    (Note: OpenSearch data: 1.5x size of raw data)
  - $$$

AWS Managed OpenSearch

- Explicitly provisioned
  - Instance types (typically r6g.*.search
  - # of nodes (instances)
  - storage (EBS)
- Scaled manually
  - Scaling down is possible with some effort

AWS Managed OpenSearch

Costs:

- ~$0.178 / r7g.large.search-hour (On-Demand)
  - reserved: up to 50% reduction
  - graviton ('g') instances: up to 40% better price-performance
- $0.122 - $0.135 / GB-month (g3/g2)
  - reductions possible: hot -> warm -> cold storage

AWS Managed Opensearch

Pros:

- Greater control of provisioned resources
- Availability of cost reducing strategies
  - reserved instances, hot/warm/cold storage

Cons:

- More monitoring and management required
  - Manual scaling
- Tendency to over-provision
- Does not respond well to traffic spikes

The world according to Gemini

(w/ annotations)

Feature	Amazon OpenSearch Serverless (AOSS)	AWS Managed OpenSearch Service
Management Overhead	Very Low. AWS handles all cluster, capacity, patching, and version upgrades non-disruptively. The only available means of intervention is via capacity limits.	Moderate. AWS manages infrastructure, node replacements, and basic cluster setup, but you manage capacity and scaling.
Scalability	Automatic and on-demand. Scales compute resources based on workload needs without manual intervention.	Manual or API-driven. You select instance types and scale by adding or changing nodes, which can be time-consuming.
Cost Model	Pay-per-use (per OCU-hour and GB-month of storage). Cost ~~varies with~~ based on actual usage, no upfront provisioning.	Provisioned capacity (per instance-hour, storage, and data transfer). Costs are predictable and can be optimized with Reserved Instances.
Best Use Case	~~Infrequent, intermittent, or unpredictable workloads~~ Frequently spiky (more spikes than not) and steadily-high workloads, broad data access w/ high access:volume ratio, and for smaller datasets/startups prioritizing simplicity.	Steady, predictable workloads that require specific configurations, plugins, or larger-scale control.

The PDS Use Case

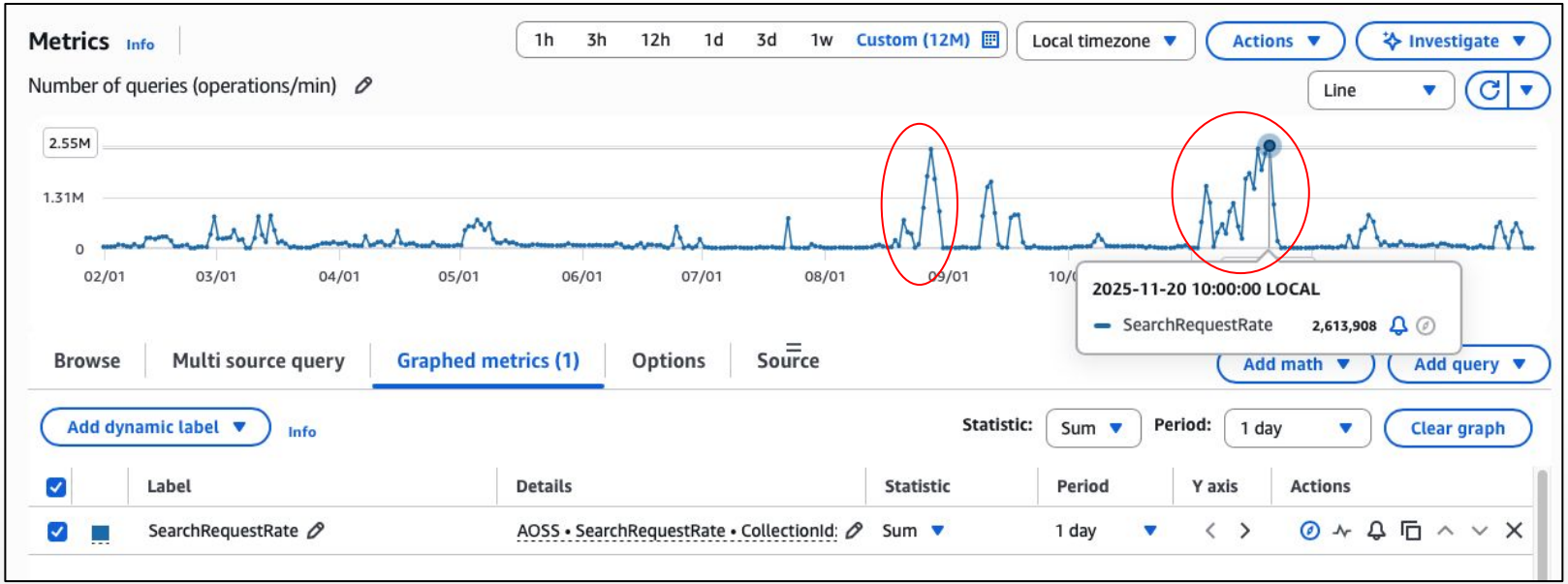
Data and graphs taken from AOSS dashboards/monitoring but demonstrate storage and search activity for any OpenSearch approach.

I assume none of the following comes as a big surprise.

pds-mcp-registry-prod

Queries:

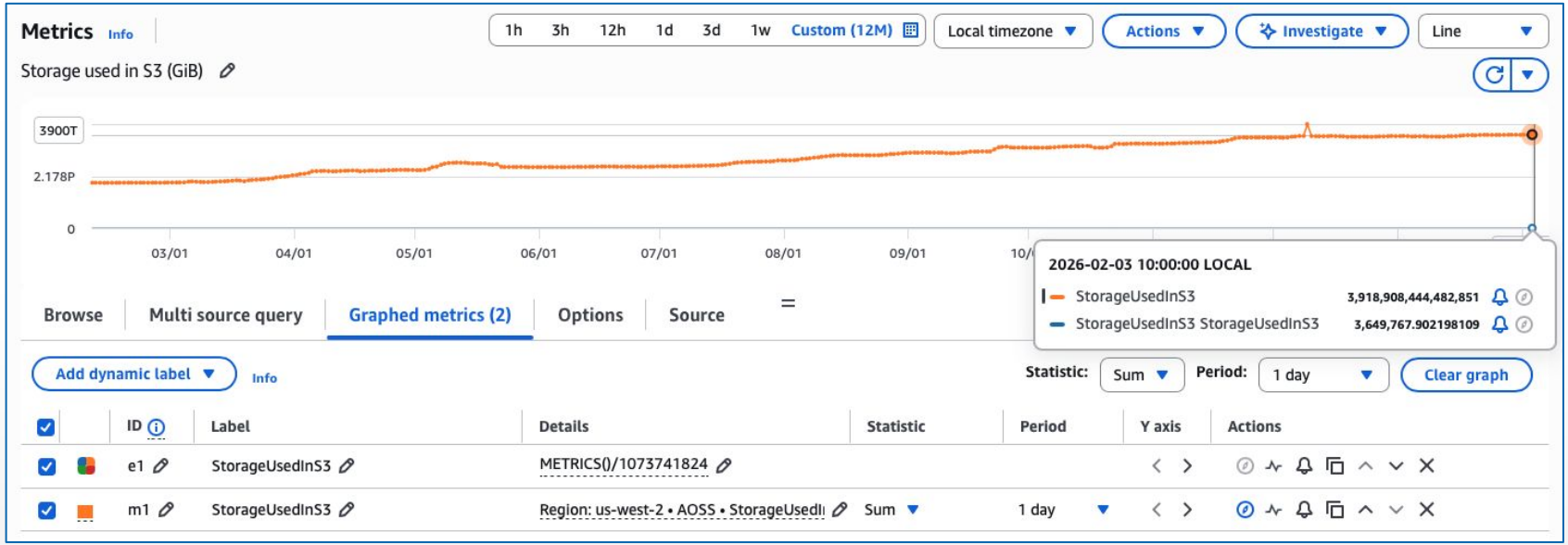
- Very spiky (even discounting the suspicious looking extremes)



pds-mcp-registry-prod

Storage:

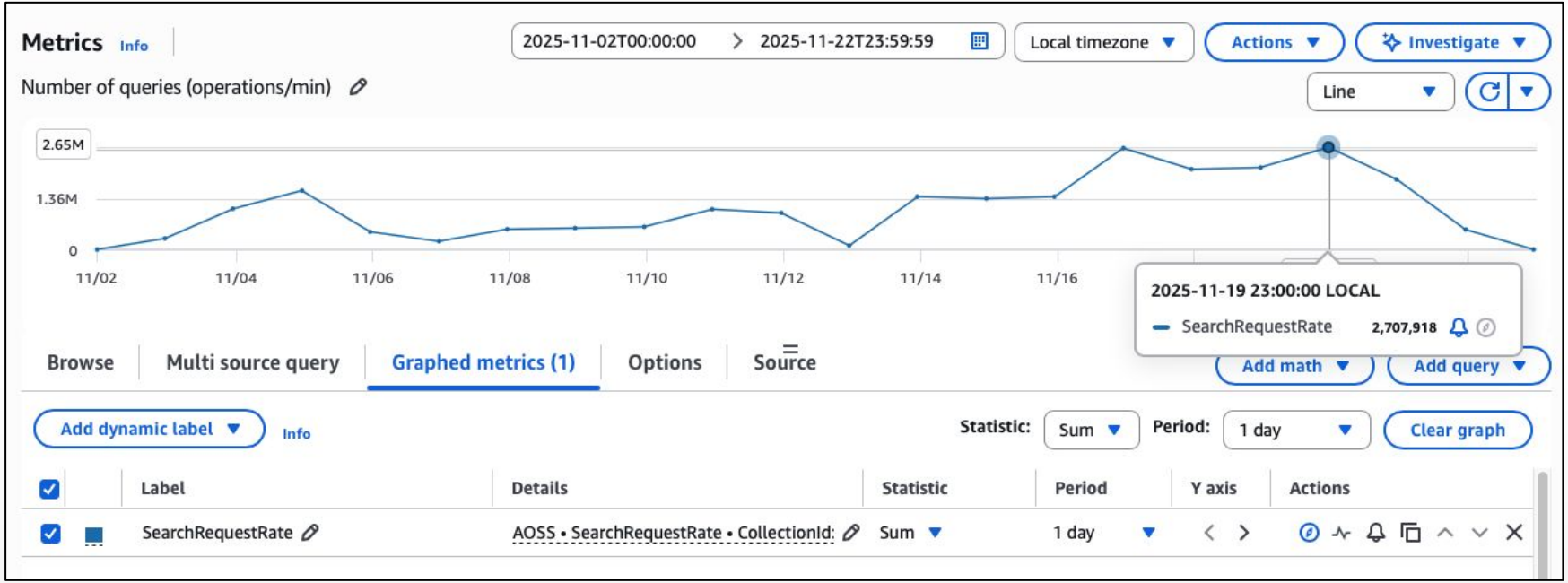
- 2.47TB (1.5x ~= 3.7TB)
- Largely steady growth



Q: What is our fore-awareness of large bulk uploads in terms of lead time, size and duration?

pds-mcp-registry-prod

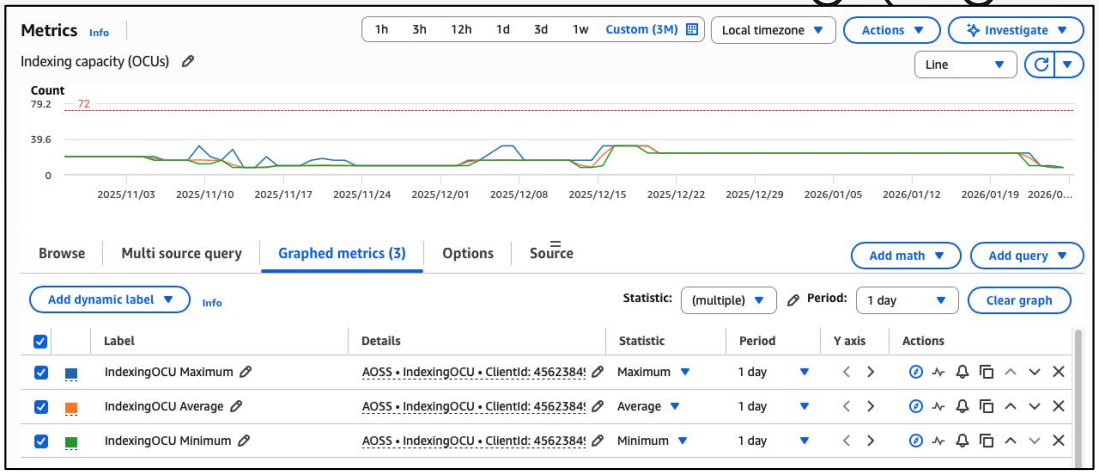
November '25

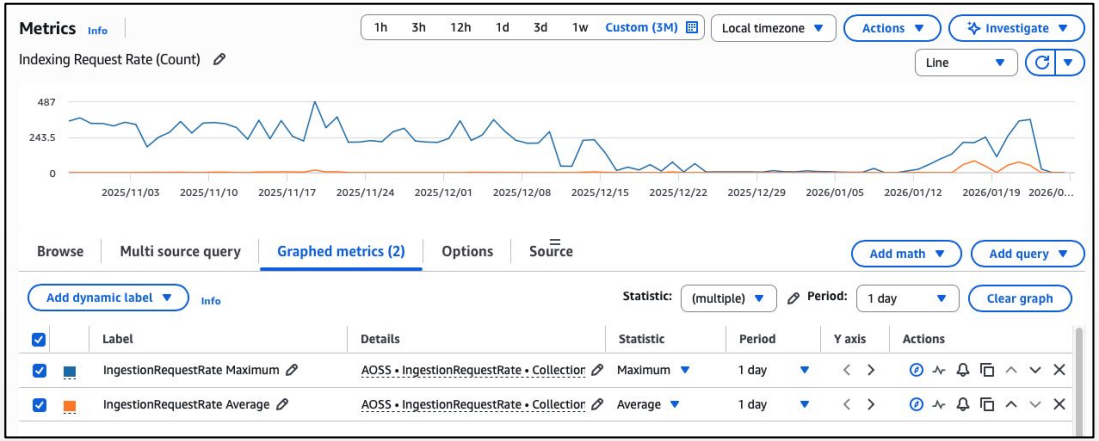


Summary Characterization

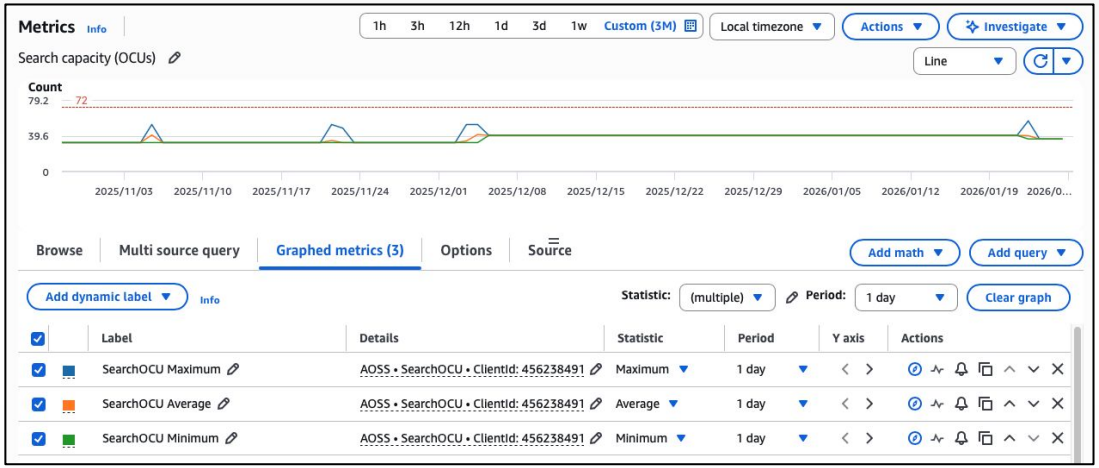
- Storage needs, while steadily increasing, are largely predictable.
  - Additional storage can be provisioned on an as needed basis.
- Search capability widely varies but must be able to accommodate the (legitimate) spikes (what about DOS?).
  - Statically provisioned compute resources will sit idle for significant periods of time.

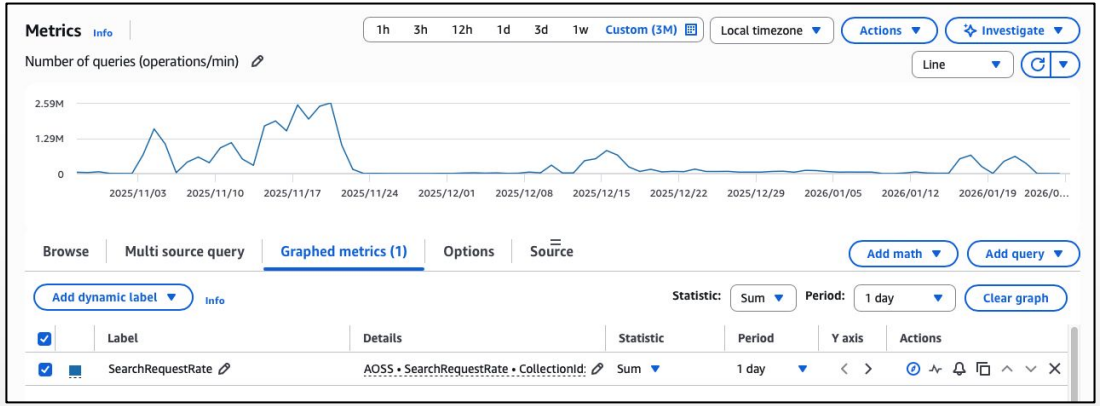
AOSS utilization - indexing (ingest)





AOSS utilization - search/storage





Cost Models (yay)

AOSS:

Function	Capacity	Rate	Monthly Cost
Search	(2.5TB / 120GB / OCU) * 2 = 44 OCUs	$0.24 OCU/hr	$7,705
Ingest	8 OCUs	$0.24 OCU/hr	$1,400
		Total / Month	$9,105

monthly = hourly * 24 hrs/day * 30.4 days/month

An interlude to answer questions from last time…

Up/Down Sizing Clusters

- Yes. Provisioned OpenSearch clusters can be both up and down scaled via the AWS Dashboard and/or `UpdateDomainConfig` API.
  - `InstanceCount`
  - `InstanceType` (be wary if using reserved instances)
  - `EBSSettings`

```
aws opensearch update-domain-config \
     --domain-name my-domain \
     --cluster-config InstanceType=r6g.xlarge,InstanceCount=4
```

AOSS Feature/Functional Gaps

- Cluster/node management (by nature)
  - _nodes, _cluster/settings, _cluster/health, _cluster/stats
- Index management and sharding APIs (also by nature)
  - _index/open, _index/close, _shrink, _split, _rollover, _settings (index level)
- Feature-specific
  - Machine learning : _train, _predict, _batch_predict, etc.
  - Snapshot APIs : manually take or restore snapshots
  - Scroll API : _search/scroll
  - _reindex

AOSS Feature/Functional Gaps

- Data Management
  - Index : state Management, rollups, transforms
- Cross-cluster replication
- Cross-region search
- Plugins
  - Anomaly detection, Alerting, k-NearestNeighbors
  - Custom plugins

ElastiCache

- Redis : Store results of frequent or computationally expensive queries
  - Depends
    - on PDS access patterns
    - ability to evaluate query equivalence
- Semantic Cache : store conversational memory
  - Value TBD but there's potential here
- Log Analytics / Monitoring
  - N/A
- Costs (typically) based on GB-hour, ECPUs (!!!)

SQS

- Ingest pipelines
  - Response to (un)expected bursts of activity
  - SQS becomes the scale enabler
  - 'Usage cost model
    - Millions of requests : first 1M free, $0.50/M to 100B
    - Egress (SQS involvement is inconsequential)
    - <u>No idle cost</u> (unlike EC2, can't be overprovisioned)
- Things to consider
  - Async model
  - Detection, re-routing and caller conveyance
    - Use by default?

End interlude

Possible Cost Optimizations

Strategy	Applicability	Notes
Reserved Instances	Yes	Reserved instances on a 1 year term reduces monthly cost by 35% ($1,134 vs $1,747). 3 year term further increases savings but w/ high upfront cost & commitment.
Graviton Instances	Yes	This has been incorporated into the previous cost model. Typical savings are 10-20% base cost and up to 30-40% performance based.
UltraWarm Storage **Writable Warm** Cold Storage	TBD	Shifts a % of read-only, less frequently accessed data to S3 and management by ultrawarm nodes. 90% movement results in 50% reduction in cost. Are PDS data access patterns sufficiently predictable to minimize performance impact?
Zstd Compression	TBD	Available as a index-level codec for v2.9 and later. Typically 30% improvement over default. Varying levels (1-6) for performance considerations. Needs to be tested further against PDS data.

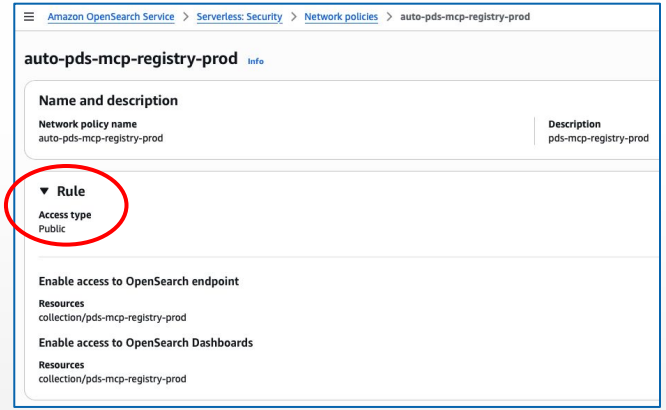
Migration

FYI: OpenSearch Ingestion Pipeline

- AOSS : source, Managed OS : sink
- stream data, no downtime (in theory)
- but … (per AWS documentation)

  "You can only migrate data to or from OpenSearch Serverless collections that use VPC access. Public collections aren't supported."

- along w/ …



Cost Models (yay)

Managed OpenSearch:

Function	Capacity	Rate	Monthly Cost
Storage	gp3 : 2.5 TB * 1.4 * 2 = 7 TB (7,168 GB)	$0.122 GB/month	$875
Data Nodes	r6g.2xlarge.search (3)	$0.67/hr	$1,467
Master Nodes	m6g.large.search (3)	$0.128/hr	$280
		Total / Month	$2,622

storage = data * 1.4 overhead * 2 replicas
monthly = hourly * 24 hrs/day * 30.4 days/month
(# of data nodes needs to be validated)

Fortunately…

- Snapshot and Restore will work
- Requires downtime (it's a comprehensive *point in time*)

Planning and Best Practices (per Gemini)

- Plan and Test: A careful planning activity and a proof-of-concept (POC) exercise are crucial to ensure the target environment meets cost, security, and performance goals. (Also, impact and necessary changes to production configs.)
- Communicate the intentions.
- **Version** Compatibility: Ensure the OpenSearch versions are compatible. Generally, the target domain's version should be the same or higher than the source. (Do we have any version restrictions and/or dependencies?)
- IAM Permissions: Proper IAM roles and access policies are critical for secure communication between services (e.g., AOSS, OpenSearch Ingestion, S3).
- **Cut Over**: After verifying data synchronization and performance, you can cut over to the new managed environment. (We will have to run both for a while w/ no ingest. Need to identify means for a comprehensive validation of all indices and a performance comparison.)

-NOPE-

Next steps

(aka, verifying that Jimmie isn't just blowing smoke)

Somewhat quick and dirty POC (2-3 weeks)

- Create plan (POC and production levels)
  - ID & test cost optimizations that are feasible/applicable
- Dev Setup
- **Identify suitable migration strategy**
- Dev Migrate data
- Deploy dev service(s) against (note config changes)
- Brief on-the-fly cost comparisons w/ sub-dataset
- Assess Performance
  - a more comprehensive assessment after production migration
- Bob = Uncle

Next steps - Prod

- Final validation of plan, POC findings
- Announcement & timeframe(s), look for nodding heads
- Provision and set up Managed OS resources
- Freeze
- Snapshot
- Restore
- Assess - integrity and performance
- Reconfigure services, test, re-open for business
- Keep a watchful eye
- Bob = Uncle again

Also…

- To keep provisioned resources reasonable:
  - Avoid DOS for legitimate searches
  - Identify legitimacy of those really big spikes
    - Slide 13
  - If not legit, black-list

Thank you

Backup

TILs

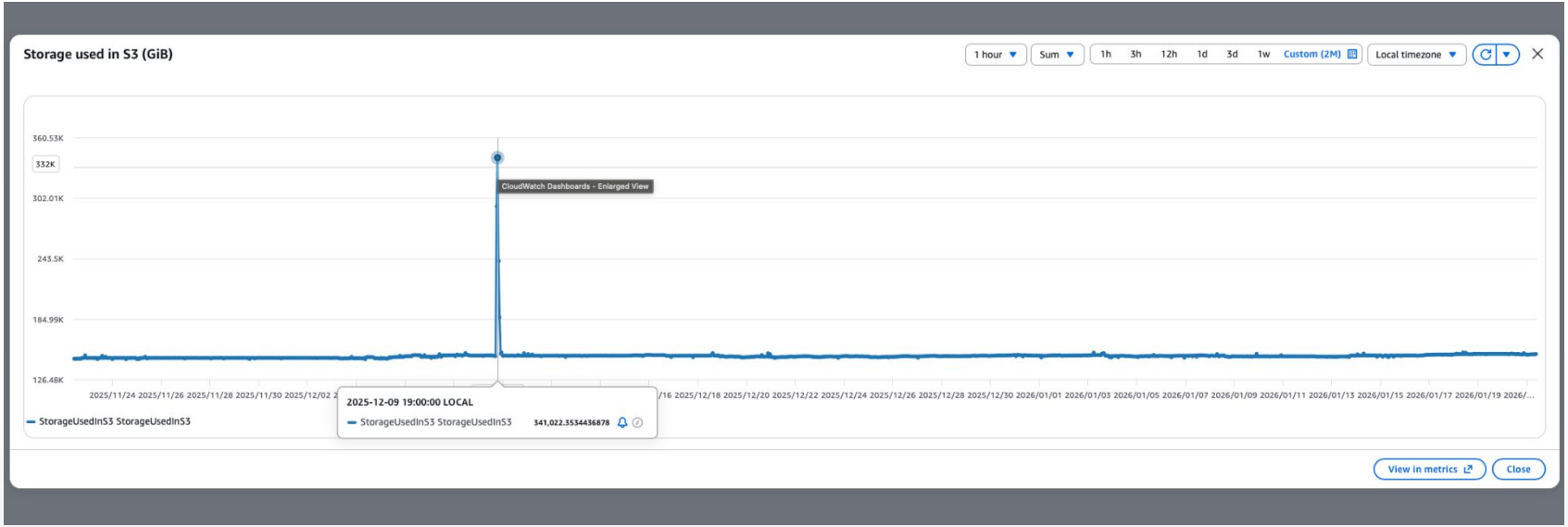
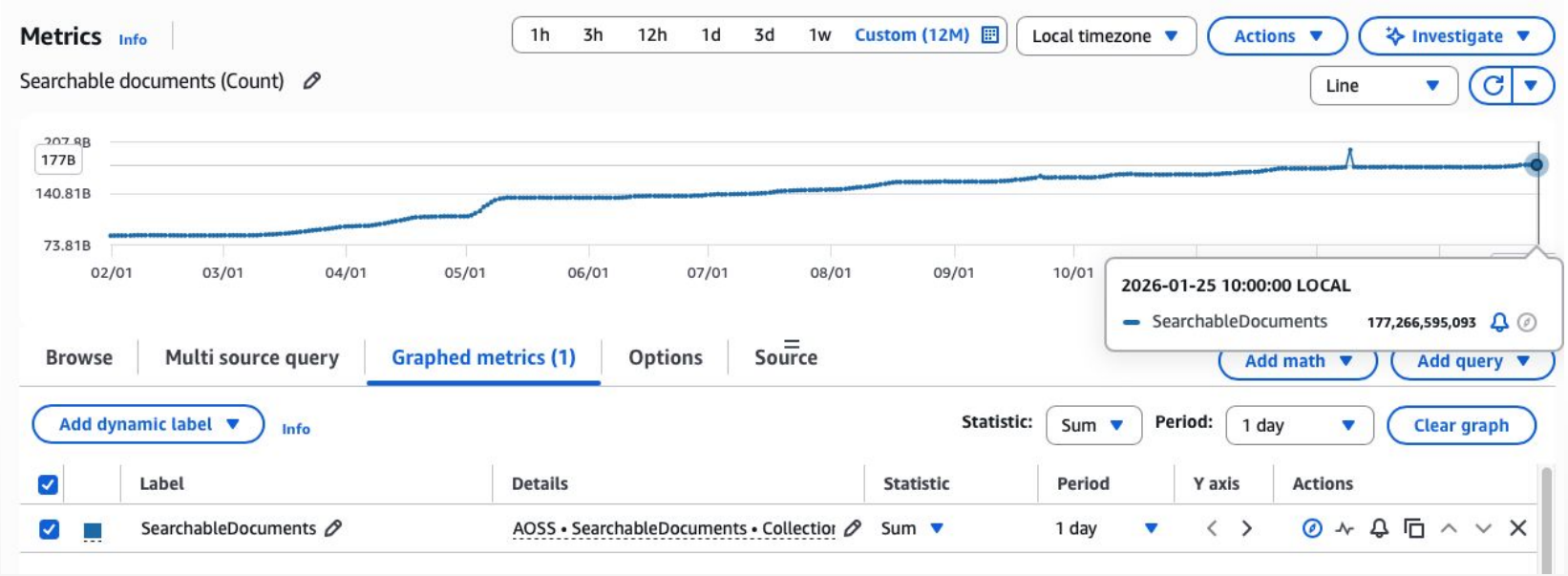
- Standard Reserved Instances can be increased in size in the same family. Convertible RI's can be increased (cost-wise) across families and sizes.

- *Reserved Instance Marketplace* where you can sell up to the lower of $50,000 or 5,000 of reserved instances.

- Vector collections : Gen AI applications involving semantic search. Distance metrics & similarities

Spike

(disregard volume numbers; i.e. it's the S3 sum() trap)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